

Taruni Chopparapu

AI/ML Engineer

FL | 904-257-3162 | taruni.c@zohomail.com | [GitHub](#) | [LinkedIn](#)

SUMMARY

AI/ML Engineer with 4+ years of experience building production-grade machine learning and LLM systems. Skilled in ML pipelines, RAG architectures, deep learning, and MLOps using Python, Scikit-learn, XGBoost, TensorFlow, LangChain, and Hugging Face. Experienced deploying scalable AI solutions on AWS with Docker, Kubernetes, SageMaker, and CI/CD, improving model accuracy, accelerating deployments, and enabling reliable data-driven applications across enterprise environments.

SKILLS

- Methodologies:** SDLC, Agile, MLOps, Model Lifecycle Management, Cross-Functional Collaboration
- Programming Languages:** Python, R, Java, SQL
- Machine Learning & Deep Learning:** TensorFlow, PyTorch, Scikit-learn, Keras, XGBoost, LightGBM, CNNs, RNNs, LSTMs, GANs, Transformers, Autoencoders
- Natural Language Processing & Generative AI:** Large Language Models (GPT, BERT, Llama), LangChain, Hugging Face Transformers, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embedding Optimization, LLM Fine-Tuning
- Computer Vision & Analytics:** OpenCV, Feature Engineering, Object Detection (YOLO, Efficient Net), Image Segmentation, Visualization (Tableau, Power BI, Seaborn, Matplotlib)
- Data Science & Engineering Tools:** Pandas, NumPy, SQL Alchemy, Feature Stores, ETL Development, Airflow, Data Wrangling, Statistical Modeling
- DevOps & Cloud Technologies:** AWS SageMaker, EC2, Lambda, Docker, Kubernetes, Git, Jenkins, CI/CD for ML, Model Versioning, Monitoring, RESTful APIs

EXPERIENCE

McKesson, FL | July 2025 - Current | AI/ML Engineer

- Designed and deployed end-to-end ML pipelines using Python, Scikit-learn, XGBoost, and LightGBM, improving model inference accuracy by 18% across clinical and operational use cases.
- Built and productionized LLM-powered RAG systems using LangChain, Hugging Face Transformers, vector embeddings, and GPT-based models, reducing knowledge retrieval latency by 35% for internal stakeholders.
- Implemented MLOps workflows with AWS SageMaker, Docker, Kubernetes, and CI/CD pipelines, cutting model deployment time from weeks to under 2 days while ensuring full model versioning and rollback support.
- Developed and fine-tuned deep learning models (CNNs and Transformers) for document intelligence and signal extraction, enabling scalable automation without compromising regulatory compliance.
- Engineered robust data pipelines and feature stores using Pandas, NumPy, SQL Alchemy, Airflow, and ETL frameworks, improving data reliability and consistency across multiple AI initiatives.
- Partnered cross-functionally with product, regulatory, and data engineering teams using Agile SDLC practices, translating business problems into production-grade AI solutions aligned with enterprise governance standards.

Virtual Infotech Solution, India | Apr 2020 - Nov 2023 | ML Engineer

- Developed supervised and ensemble ML models using Scikit-learn, XGBoost, and LightGBM for customer behavior and risk prediction, improving model precision by 18% across client deployments.
- Designed ETL pipelines with Pandas, SQL Alchemy, and Airflow to process large-scale structured and unstructured datasets, reducing data preprocessing time by 40%.
- Built deep learning models including CNNs and LSTMs, using TensorFlow and Keras for time-series forecasting and image classification, achieving up to 25% improvement in prediction stability.
- Implemented feature engineering and statistical modeling techniques that improved downstream model performance and reduced feature noise without increasing model complexity.
- Deployed ML inference services using Dockerized REST APIs, enabling scalable model consumption by frontend and backend systems across multiple projects.
- Created data visualizations and analytical dashboards using Matplotlib, Seaborn, Tableau, and Power BI to communicate insights clearly to non-technical stakeholders and decision makers.
- Worked closely with cross-functional teams to translate business requirements into ML solutions, contributing to repeat client engagements and long-term platform enhancements.

EDUCATION

Master of Science in information systems and operations management	Dec 2025
University of Florida, Gainesville, Florida, USA	
Bachelor of Technology in Computer Science	Apr 2022
Jawaharlal Nehru Technological University, Kakinada, Andhra Pradesh, India	